# MA1503 Final Exam Tutorial Solution

1. (1) A function with $f(u+v) = f(u) + f(v)$ and $f(cv) = cf(v)$ for all $u, v \in \mathbb{R}^n$ and $c \in \mathbb{R}$.

   (2) 0, 1, or infinitely many.

   (3) A subset $V$ of a vector space is a subspace if it contains the zero vector such that if $u, v \in V$ and $c \in \mathbb{R}$, then $u + v \in V$ and $cv \in V$.

   (4) Row reduce $A$ and count the number of pivot columns.

   (5) Row reduce the $n \times 2n$ matrix $\begin{bmatrix} A & I \end{bmatrix}$ and then take the right $n \times n$ submatrix of the result.

   (6) One answer is the parallelogram with sides $\begin{bmatrix} a \\ c \end{bmatrix}$ and $\begin{bmatrix} b \\ d \end{bmatrix}$.

   (7) $A^T Ax = A^T b$.

   (8) Symmetric matrices.

   (9) The square roots of the eigenvalues of $A^T A$.

   (10) The shape is an ellipse, centered at the origin, with radius vectors whose lengths are the diagonal entries of $\Sigma$, and whose directions are given by the columns of $U$.

2. (1) Suppose $\lambda = 0$. Let $v$ be a 0-eigenvector so that $Av = 0$. Since $v \neq 0$, this implies that the columns of $A$ are linearly dependent. By the invertible matrix theorem, $A$ is not invertible, which is a contradiction. Thus, $\lambda \neq 0$.

   (2) Let $v$ be a $\lambda$-eigenvector of $A$, so $Av = \lambda v$. Since $A$ is invertible and $\lambda \neq 0$, we have:

   $$A^{-1}(Av) = A^{-1}(\lambda v) \implies v = \lambda A^{-1} v \implies A^{-1} v = \lambda^{-1} v$$

   Therefore, $\lambda^{-1}$ is an eigenvalue of $A^{-1}$.

3. (1) Note that $A^T - \lambda I = (A - \lambda I)^T$. Therefore,

   $$p_{A^T}(\lambda) = \det((A - \lambda I)^T) = \det(A - \lambda I) = p_A(\lambda)$$

   since $\det(B) = \det(B^T)$ for any $B$.
   Since $A$ and $A^T$ have the same characteristic polynomial, they have the same set of eigenvalues.

   (2) Suppose $\lambda$ is an eigenvalue of $A$. Then $Av = \lambda v$ for some $v \neq 0$.

   $$\implies A^2 v = A(\lambda v) = \lambda A v = \lambda(\lambda v) = \lambda^2 v.$$

   $$\implies (A^2 + I)v = A^2 v + v = \lambda^2 v + v = (\lambda^2 + 1)v.$$

   So $\lambda^2 + 1$ is an eigenvalue of $A^2 + I$.

4. (1) $E_3$ is a subspace of $\mathbb{P}_3$ if it satisfies three conditions: the zero polynomial is in $E_3$, it is closed under addition, and it is closed under scalar multiplication.

(2) Let $z(x) = 0$. Since $z(x) = 0 = z(-x)$, the zero polynomial is in $E_3$. Let $p, q \in E_3$ and $c \in \mathbb{R}$. Then $(p+q)(-x) = p(-x) + q(-x) = p(x) + q(x) = (p+q)(x)$, so $p+q \in E_3$.
Also, $(cp)(-x) = c(p(-x)) = c(p(x)) = (cp)(x)$, so $cp \in E_3$.
Thus, $E_3$ is a subspace.

(3) A general polynomial in $\mathbb{P}_3$ is $p(x) = a_0 + a_1x + a_2x^2 + a_3x^3$. For $p(x) = p(-x)$, we must have $a_1x + a_3x^3 = -a_1x - a_3x^3$ for all $x$, which implies $a_1 = a_3 = 0$.
Thus, $p(x) = a_0 + a_2x^2$.
A basis for $E_3$ is $\{1, x^2\}$.

5. To show that no such invertible matrix $C$ exists, let $\mathbf{a}_1, \ldots, \mathbf{a}_4$ and $\mathbf{b}_1, \ldots, \mathbf{b}_4$ be the columns of $A$ and $B$, respectively. Define the $3 \times 3$ submatrices as $A' = [\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3]$ and $B' = [\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3]$.

**Step 1: Linear dependence of the columns of $B'$.**
The submatrix $B'$ is given by:

$$B' = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ 0 & b_{22} & b_{23} \\ 0 & 0 & 0 \end{bmatrix}$$

Since the third row of $B'$ consists entirely of zeros, its row rank is at most 2. Consequently, its column rank is also at most 2. Since the number of columns (3) is greater than the rank, the columns $\{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}$ are **linearly dependent**. Thus, there exists a non-zero vector $\mathbf{k} \neq \mathbf{0}$ such that:

$$B'\mathbf{k} = \mathbf{0}$$

**Step 2: Linear independence of the columns of $A'$.**
The submatrix $A'$ is given by:

$$A' = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{bmatrix}$$

Given that $a_{11}, a_{22}, a_{33} \neq 0$, $A'$ is an upper triangular matrix with non-zero diagonal entries. Its determinant $\det(A') = a_{11}a_{22}a_{33} \neq 0$, which implies that the columns $\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$ are **linearly independent**.

**Step 3: Contradiction via the matrix equation.**
Suppose there exists an invertible matrix $C$ such that $A = CB$. This implies that for the first three columns, $A' = CB'$. Multiplying by the non-zero vector $\mathbf{k}$ from Step 1:

$$A'\mathbf{k} = (CB')\mathbf{k} = C(B'\mathbf{k})$$

Substituting $B'\mathbf{k} = \mathbf{0}$, we obtain:

$$A'\mathbf{k} = C \cdot \mathbf{0} = \mathbf{0}$$

However, since the columns of $A'$ are linearly independent, the equation $A'\mathbf{x} = \mathbf{0}$ has only the trivial solution $\mathbf{x} = \mathbf{0}$. This contradicts our finding that $\mathbf{k} \neq \mathbf{0}$.

**Conclusion:**
The assumption leads to a contradiction. Therefore, no such invertible matrix $C$ exists.

6. (1) The complex conjugate $1 + i$ must also be an eigenvalue of $A$. So $A$ has three distinct eigenvalues, is diagonalizable, and can be written as

$$A = P \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1-i & 0 \\ 0 & 0 & 1+i \end{bmatrix} P^{-1}.$$

Therefore,

$$A^{-1} = P \begin{bmatrix} 2^{-1} & 0 & 0 \\ 0 & (1-i)^{-1} & 0 \\ 0 & 0 & (1+i)^{-1} \end{bmatrix} P^{-1} = P \begin{bmatrix} 1/2 & 0 & 0 \\ 0 & (1+i)/2 & 0 \\ 0 & 0 & (1-i)/2 \end{bmatrix} P^{-1}.$$

And so

$$A^{-1} - xI = P \begin{bmatrix} 1/2 - x & 0 & 0 \\ 0 & (1+i)/2 - x & 0 \\ 0 & 0 & (1-i)/2 - x \end{bmatrix} P^{-1}.$$

Hence

$$f(x) = \det(A^{-1} - xI) = (1/2 - x)\left(\frac{1+i}{2} - x\right)\left(\frac{1-i}{2} - x\right) = (1/2 - x)\left(\frac{1}{2} - x + x^2\right).$$

(2)

$$f(5) = \left(\frac{1}{2} - 5\right)\left(\frac{1}{2} - 5 + 25\right) = \left(-\frac{9}{2}\right)\left(\frac{41}{2}\right) = -\frac{369}{4}.$$

7. (1) No. Any such matrix $A$ has 3 linearly independent eigenvectors so is diagonalizable.

   (2) No. A matrix is non-invertible if and only if it has 0 as an eigenvalue, but $A$ only has $-1$ and 2 as eigenvalues.

   (3) No. Orthogonal matrices define length-preserving linear transformations, so can only have $\pm 1$ as eigenvalues. But $A$ has 2 as an eigenvalue.

   (4) Yes.

   Suppose $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$ are in the $-1$-eigenspace of $A$, while $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ is in the 2-eigenspace.

   Note that $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$ are not orthogonal since their dot product is $1 \neq 0$.

   To construct an orthonormal basis, we apply the Gram-Schmidt process to these two vectors in the $-1$-eigenspace.

   Let $\mathbf{v}_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$. Then

$$\mathbf{v}_2 = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} - \text{proj}_{\mathbf{v}_1} \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} - \frac{\begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} - \frac{1}{1} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \\ 0 \end{bmatrix}.$$

   Normalizing these vectors (and the third eigenvector) gives the orthonormal basis:

$$\mathbf{u}_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{u}_2 = \frac{1}{\sqrt{8}} \begin{bmatrix} 2 \\ -2 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}, \quad \mathbf{u}_3 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}.$$

   With respect to this orthonormal basis, $A$ is orthogonally diagonalizable:

$$A = \begin{bmatrix} 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & -1/\sqrt{2} & 1/\sqrt{2} \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \end{bmatrix}.$$

Since $A = QDQ^T$ where $Q$ is orthogonal, the matrix $A$ is symmetric.

8. From the given information, we know that $A = PDP^{-1}$ for

$$P = \begin{bmatrix} 2 & 2 \\ 2 & -2 \end{bmatrix} \quad \text{and} \quad D = \begin{bmatrix} \lambda & 0 \\ 0 & 1 \end{bmatrix}$$

Then the inverse of $P$ is calculated as:

$$P^{-1} = \frac{1}{-8} \begin{bmatrix} -2 & -2 \\ -2 & 2 \end{bmatrix} = \frac{1}{8} \begin{bmatrix} 2 & 2 \\ 2 & -2 \end{bmatrix}$$

Thus, matrix $A$ is:

$$A = \frac{1}{8} \begin{bmatrix} 2 & 2 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} \lambda & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 2 & -2 \end{bmatrix}$$

To compute $\lim_{n\to\infty} A^n$, we use the property $A^n = PD^nP^{-1}$. Since $-1 < \lambda < 1$, we have $\lim_{n\to\infty} \lambda^n = 0$. Therefore:

$$\lim_{n\to\infty} D^n = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

Finally, the limit of $A^n$ is:

$$\lim_{n\to\infty} A^n = \frac{1}{8} \begin{bmatrix} 2 & 2 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 2 & -2 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

9. (1) Any square zero matrix will do, such as the $1 \times 1$ matrix $\begin{bmatrix} 0 \end{bmatrix}$.
   (2) An $n \times n$ triangular matrix with all distinct diagonal entries has $n$ distinct eigenvalues so is diagonalizable. So one example is $\begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$.
   (3) Any triangular-but-not-diagonal square matrix with equal diagonal entries is not diagonalizable. So any example is $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$.
   (4) Every permutation matrix is invertible, but only the identity matrix is triangular. So one example is $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.
   (5) The matrix $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$ is orthogonal: its columns are orthonormal. It is not a rotation matrix since every rotation matrix $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ has equal diagonal entries.

10. (1) $A$ is diagonalizable. Since $A$ is a real symmetric $3 \times 3$ matrix, it follows from the Spectral Theorem that it is orthogonally diagonalizable over $\mathbb{R}$.
    (2) The eigenvalues are $\lambda_1 = 1$ and $\lambda_2 = 2$ from $c_A(\lambda) = (1-\lambda)(2-\lambda)^2$.
    - For $\lambda_1 = 1$: Algebraic multiplicity is 1, so geometric multiplicity $d_1 = 1$. A basis for $E_1(A)$ is:

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \right\}$$

- For $\lambda_2 = 2$: Algebraic multiplicity is 2, so geometric multiplicity $d_2 = 2$.
- Since $A$ is symmetric, $E_2(A)$ must be the orthogonal complement of $E_1(A)$. The basis vectors must satisfy $x + y - z = 0$. A basis for $E_2(A)$ is:

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$$

(3) **Matrix $A$:**
Using $A = PDP^{-1}$ with the eigenvectors as columns of $P$:

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ -1 & 1 & 1 \end{bmatrix}^{-1}$$

11. An orthogonal basis for $V = \mathbb{R}\text{-span}\{v, w\}$ is

$$v = \begin{bmatrix} 1 \\ 1 \\ -1 \\ 0 \end{bmatrix} \quad \text{and} \quad u = 3\left(w - \frac{w \bullet v}{v \bullet v} v\right) = 3\left(\begin{bmatrix} -1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + \frac{1}{3}\begin{bmatrix} 1 \\ 1 \\ -1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} -2 \\ 4 \\ 2 \\ 3 \end{bmatrix}.$$

The vector in $V$ that is as close to $x$ as possible is

$$\text{proj}_V(x) = \frac{x \bullet v}{v \bullet v} v + \frac{x \bullet u}{u \bullet u} u = \begin{bmatrix} v & u \end{bmatrix} \begin{bmatrix} \frac{v \bullet x}{v \bullet v} \\ \frac{u \bullet x}{u \bullet u} \end{bmatrix} = \begin{bmatrix} v & u \end{bmatrix} \begin{bmatrix} \frac{1}{v \bullet v} & 0 \\ 0 & \frac{1}{u \bullet u} \end{bmatrix} \begin{bmatrix} v^\top \\ u^\top \end{bmatrix} x.$$

As $v \bullet v = 3$ and $u \bullet u = 4 + 16 + 4 + 9 = 33$ the desired matrix is

$$A = \begin{bmatrix} 1 & -2 \\ 1 & 4 \\ -1 & 2 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1/3 & 0 \\ 0 & 1/33 \end{bmatrix} \begin{bmatrix} 1 & 1 & -1 & 0 \\ -2 & 4 & 2 & 3 \end{bmatrix}$$

12. **(1)** Two linearly independent vectors in $P$ are $\begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 6 \\ 1 \end{bmatrix}$. The Gram-Schmidt process converts the second vector to

$$\begin{bmatrix} 0 \\ 6 \\ 1 \end{bmatrix} - \frac{18}{10}\begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 6 \\ 1 \end{bmatrix} - \begin{bmatrix} 9/5 \\ 27/5 \\ 0 \end{bmatrix} = \begin{bmatrix} -9/5 \\ 3/5 \\ 1 \end{bmatrix}.$$

After rescaling this vector, we get an orthogonal basis for $P$ given by

$$\left\{ \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} -9 \\ 3 \\ 5 \end{bmatrix} \right\}.$$

**(2)** The desired vector is the projection of $v$ onto $P$ which is

$$\frac{\begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} \bullet \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}}{\begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix} \bullet \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}} \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix} + \frac{\begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} \bullet \begin{bmatrix} -9 \\ 3 \\ 5 \end{bmatrix}}{\begin{bmatrix} -9 \\ 3 \\ 5 \end{bmatrix} \bullet \begin{bmatrix} -9 \\ 3 \\ 5 \end{bmatrix}} \begin{bmatrix} -9 \\ 3 \\ 5 \end{bmatrix}$$

which we can rewrite as

$$\frac{9}{10}\begin{bmatrix}1\\3\\0\end{bmatrix}+\frac{-16}{115}\begin{bmatrix}-9\\3\\5\end{bmatrix}=\frac{1}{230}\left(207\begin{bmatrix}1\\3\\0\end{bmatrix}-32\begin{bmatrix}-9\\3\\5\end{bmatrix}\right)$$
$$=\frac{1}{230}\begin{bmatrix}495\\525\\-160\end{bmatrix}$$
$$=\frac{1}{46}\begin{bmatrix}99\\105\\-32\end{bmatrix}.$$

13. (1) The augmented matrix of the system $Ax=b$ is

$$\left[\begin{array}{cc|c}1&3&2\\0&-1&1\\2&2&2\end{array}\right]$$

which has reduced echelon form

$$\left[\begin{array}{cc|c}1&0&0\\0&1&0\\0&0&1\end{array}\right].$$

Since the last column is a pivot column, the original linear system is inconsistent so has no exact solution.

(2) The least-squares solutions to $Ax=b$ are the solutions to $A^\top Ax=A^\top b$. Since

$$A^\top A=\begin{bmatrix}5&7\\7&14\end{bmatrix}\quad\text{and}\quad A^\top b=\begin{bmatrix}6\\9\end{bmatrix}$$

the augmented matrix of $A^\top Ax=A^\top b$ is

$$\left[\begin{array}{cc|c}5&7&6\\7&14&9\end{array}\right]$$

which has reduced echelon form

$$\left[\begin{array}{cc|c}1&0&1\\0&1&1/7\end{array}\right].$$

Hence the unique least-squares solution to $Ax=b$ is $x=\begin{bmatrix}1\\1/7\end{bmatrix}$.

(3) (a) FALSE
(b) TRUE
(c) FALSE
(d) TRUE
(e) TRUE

14. **(1)** The singular values of $A$ are the square roots of the eigenvalues of

$$A^\top A=\begin{bmatrix}9&-9\\-9&9\end{bmatrix}.$$

We have

$$\det(A^\top A - xI) = \det \begin{bmatrix} 9-x & -9 \\ -9 & 9-x \end{bmatrix} = (9-x)^2 - 9^2 = (-x)(18-x)$$

so the eigenvalues of $A^\top A$ are $\lambda = 0$ and $\lambda = 18$. Hence the singular values are

$$\sigma_1 = \sqrt{18} > \sigma_2 = 0.$$

**(2)** The correct answer is a picture of a solid ellipse centered at the origin with radius vectors $\pm \begin{bmatrix} -8 \\ 6 \end{bmatrix}$ and $\pm \begin{bmatrix} 3 \\ 4 \end{bmatrix}$.

**(3)** A basis for the orthogonal complement of the span of this vector is

$$\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}.$$

Performing the Gram-Schmidt process converts the second vector to

$$\begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} - \frac{-4}{5} \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} -2 \\ 4 \\ 5 \end{bmatrix}$$

so an orthogonal basis of our original orthogonal complement is

$$\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 4 \\ 5 \end{bmatrix}.$$

Converting these to unit vectors gives the desired orthonormal basis:

$$\left\{ \begin{bmatrix} 1/3 \\ -2/3 \\ 2/3 \end{bmatrix}, \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \frac{1}{\sqrt{45}} \begin{bmatrix} -2 \\ 4 \\ 5 \end{bmatrix} \right\}.$$

15. (1) A shortcut to solving this part is to realize that $A$ and $A^\top$ have the same nonzero singular values, that is, the nonzero singular values of $A$ are the square roots of the eigenvalues of $AA^\top$. We have

$$AA^\top = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{bmatrix} = \begin{bmatrix} 17 & 8 \\ 8 & 17 \end{bmatrix}$$

so $\det(AA^\top - xI) = (17-x)^2 - 8^2 = (25-x)(9-x)$ has roots $\lambda_1 = 25$ and $\lambda_2 = 9$. So that first two singular values of $A$ are $\sigma_1 = 5$ and $\sigma_2 = 3$ while the third is $\sigma_3 = 0$ since $A$ has rank two.

(2) Since $V^\top$ is invertible, $\{V^\top x : x \in \mathbb{R}^3\} = \mathbb{R}^3$, so

$$\begin{aligned} \mathrm{Col}(A) &= \{U\Sigma V^\top x : x \in \mathbb{R}^3\} \\ &= \{U\Sigma x : x \in \mathbb{R}^3\} \\ &= \left\{ U \begin{bmatrix} 71x_1 \\ 72x_2 \\ 0 \end{bmatrix} : x_1, x_2 \in \mathbb{R} \right\} = \left\{ U \begin{bmatrix} x_1 \\ x_2 \\ 0 \end{bmatrix} : x_1, x_2 \in \mathbb{R} \right\} \\ &= \left\{ x_1 \begin{bmatrix} u_1 \\ u_4 \\ u_7 \end{bmatrix} + x_2 \begin{bmatrix} u_2 \\ u_5 \\ u_8 \end{bmatrix} : x_1, x_2 \in \mathbb{R} \right\} = \mathbb{R}\text{-span} \left\{ \begin{bmatrix} u_1 \\ u_4 \\ u_7 \end{bmatrix}, \begin{bmatrix} u_2 \\ u_5 \\ u_8 \end{bmatrix} \right\}. \end{aligned}$$

Thus $\begin{bmatrix} u_1 \\ u_4 \\ u_7 \end{bmatrix}$, $\begin{bmatrix} u_2 \\ u_5 \\ u_8 \end{bmatrix}$ is a basis for $\mathrm{Col}(A)$. The null space of $A$ has dimension $3 - \mathrm{rank}(A) = 1$.

A nonzero vector in $\mathrm{Nul}(A)$ is $\begin{bmatrix} v_7 \\ v_8 \\ v_9 \end{bmatrix}$ since

$$A \begin{bmatrix} v_7 \\ v_8 \\ v_9 \end{bmatrix} = U\Sigma V^\top \begin{bmatrix} v_7 \\ v_8 \\ v_9 \end{bmatrix} = U\Sigma \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = U \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = 0$$

and this vector provides a basis.

(3) The matrix $A$ transforms the unit vector $\begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}$ to the length 20 vector $\begin{bmatrix} 0 \\ 20 \end{bmatrix}$.

The region $\{Av : v \in \mathbb{D}^2\}$ is an ellipse whose radii have lengths 22 and 20. It follows that the shorter radii of this ellipse are the vectors $\begin{bmatrix} 0 \\ 20 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ -20 \end{bmatrix}$ while the longer radii are $\begin{bmatrix} 22 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} -22 \\ 0 \end{bmatrix}$.

The unit vector $\begin{bmatrix} 4/5 \\ -3/5 \end{bmatrix}$ must be mapped by $A$ to one of the latter two vectors. Therefore $A$ must have one of the two singular value decompositions

$$A = \begin{bmatrix} \pm 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 22 & 0 \\ 0 & 20 \end{bmatrix} \begin{bmatrix} 4/5 & 3/5 \\ -3/5 & 4/5 \end{bmatrix}^\top$$

which simplifies to

$$A = \begin{bmatrix} 88/5 & -66/5 \\ 60/5 & 80/5 \end{bmatrix} \quad \text{or} \quad A = \begin{bmatrix} -88/5 & 66/5 \\ 60/5 & 80/5 \end{bmatrix}.$$